

Real-Time Emotion Classification

using Artificial Neural Networks



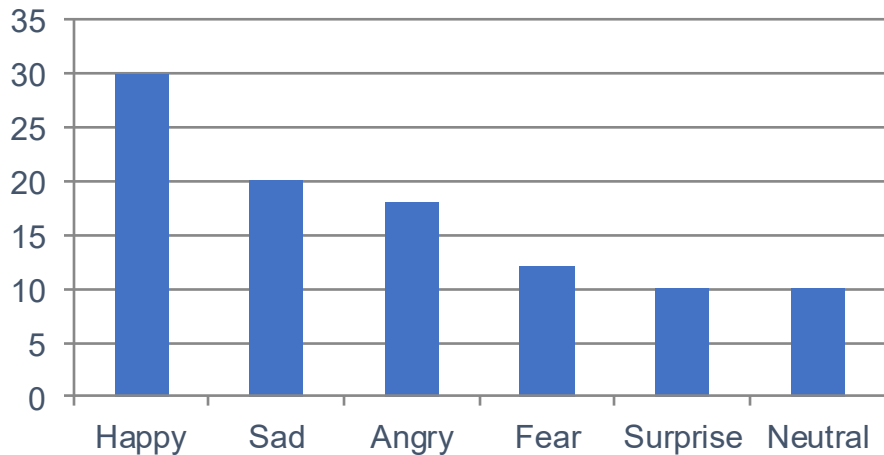
COURSE WORK: Artificial Neural Network
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Index: COMScDS242P-001

Introduction & Objectives



- **Design, train and deploy a neural network for real-time emotion recognition**
- Leverage 43,756 grayscale images across six emotion categories
- Applications: mental health, robotics, interactive agents, user feedback

Dataset Preparation



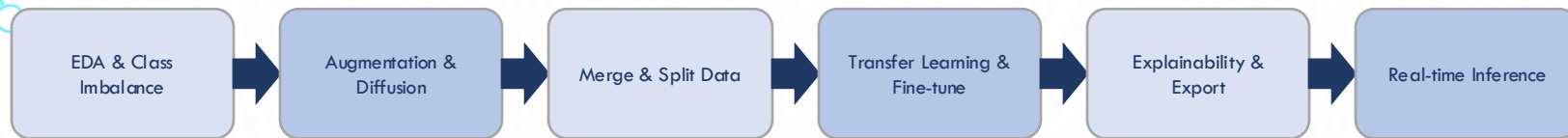
43,756 grayscale face images

6 emotion categories with significant class imbalance (Happy $\approx$ 30%)

Images resized to 48×48 & channels duplicated to RGB

Normalization & augmentation for balancing

Solution Pipeline



Sequential pipeline from data exploration to deployment.

Balancing via augmentation & diffusion improves minority class representation.

Transfer learning with fine-tuning yields high accuracy and F1 scores.

Network Architecture



ResNet50 backbone leveraged for feature extraction; custom dense head with dropout reduces overfitting.

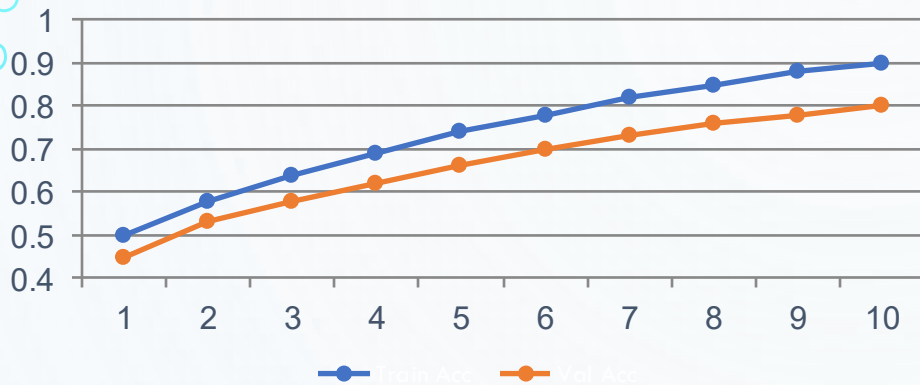
Grayscale images duplicated to RGB to match pretrained weights.

Alternative Approaches

Method	Description	Performance
Classical	SVM, HOG features	~40-50% accuracy
Scratch CNN	Small CNN trained from scratch	~60% accuracy
Transfer learning	ResNet50 with fine-tuning	70-80% accuracy
Synthetic aug.	Diffusion-generated data	Further boosts minority recall

Transfer learning and synthetic augmentation yield the highest performance and robust generalization.

Model Development & Performance



After augmenting to ~83k images and fine-tuning in two phases:

- Macro F1 score ≈ 0.813
- Stable convergence and minimal overfitting
- Minority classes see significant recall improvements

Optimisation & Hyperparameters

SGD	Stable convergence, simple	Sensitive to learning rate
AdaGrad	Adaptive step sizes	Learning rate decays too much
RMSProp	Handles non-stationary targets	Requires tuning
Adam	Combines advantages of AdaGrad and RMSProp	Can overshoot minima

Learning rate: backbone $1e-4$, head $1e-3$

Batch size: 32

Momentum & weight decay tuned via experimentation

Mobile Application



Real-time Detection

Camera feed with face detection and on-device ONNX inference (~15–20 FPS)



Photo Analysis

Upload a photo to classify emotion instantly



Image Generation

Diffusion model generates expressive faces



XAI Overlay

Grad-CAM & SHAP user trust

Flutter front-end integrates with ONNX or remote FastAPI server.

Modular tabs allow detection, analysis and generation; includes intuitive UI.

Testing & Validation

Type	Goal
Unit	Validate individual functions & data pipelines
Widget	Ensure UI elements behave correctly
Integration	Test interactions between components
Performance	Measure FPS & inference latency
User Acceptance	Gather feedback from real users

Cross-platform testing across Android, iOS & web.

Continuous integration ensures stability at each development stage.

Explainable AI

Technique	Pros	Cons
Grad-CAM	Highlights important regions	Low resolution & biased to last layers
Grad-CAM++	Better localization than Grad-CAM	Higher computational cost
LIME	Model-agnostic & interpretable locally	Sensitive to perturbations
SHAP	Consistent feature importance	Expensive for deep models

Interactive overlay allows users to toggle explanations and compare methods.

Enhances transparency and trust in emotion predictions.

Generative AI & Transformers



Self-Attention & Multi-Head Attention:

Tokens attend to every other token with learned weights. Multi-head attention runs several attention mechanisms in parallel to capture diverse relationships.

Diffusion models generate ~2,634 images to augment minority classes.



Generative vs Traditional Models

Data Dependency	Requires labeled data	Uses unsupervised / self-supervised
Expressive Power	Limited to training distribution	Can generate novel examples
Compute Needs	Moderate GPU for inference	High compute during training
Interpretability	Relatively straightforward	Less interpretable, needs XAI
Safety & Ethics	Lower risk	Misuse risk & biases

Generative methods complement discriminative neural networks by enriching data, enabling cross-modal tasks and fostering creativity.

Conclusion & Impact



A robust real-time emotion classifier built on transfer learning achieves macro F1 $\approx$ 0.813.

Open-source Flutter app enables accessible emotion recognition and explainability on mobile devices.

Generative augmentation and XAI techniques improve minority class recall and build user trust.

Future work: cross-modal emotion recognition, personalised feedback systems, ethical considerations.